

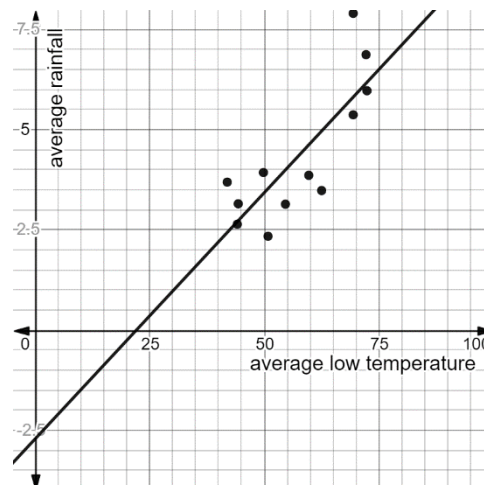
Algebra 1
Unit 3
Lesson 7 Homework

Name _____
Date _____
Period _____

A table shows the average low temperature and the average rainfall each month for a year in Jacksonville, Florida.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Average Low Temperature (°F)	41.9	44.3	49.8	54.6	62.5	69.4	72.4	72.2	69.4	59.7	50.8	44.1
Average Rainfall (in)	3.69	3.15	3.93	3.14	3.48	5.37	5.97	6.87	7.9	3.86	2.34	2.64

The scatter plot shown represents the data from the table with a linear equation fit to model the equation.



This screenshot shows the slope, y -intercept, and correlation coefficient associated with the linear relationship modeled on the scatter plot.

$$y_1 \sim mx_1 + b$$

STATISTICS

$$r^2 = 0.6461$$

$$r = 0.8038$$

PARAMETERS

$$m = 0.122798 \quad b = -2.71046$$

RESIDUALS

e_1

1. Write the equation for the line of best fit shown.

2. Interpret the meaning of the slope in terms of the context.
3. Interpret the meaning of the y -intercept in terms of the context.
4. Interpret the meaning of the correlation coefficient in terms of the context.
5. Use an inequality to express an appropriate domain in terms of the context.
6. Use an inequality to express an appropriate range in terms of the context.
7. Use the model to predict the average rainfall in a month with an average low temperature of 74°F .
8. Use the model to predict the average rainfall in a month with an average low temperature of 53°F .
9. Use the model to predict the average temperature in a month with an average rainfall of 2 inches.
10. Use the model to predict the average temperature in a month with an average rainfall of 6 inches.